N8N AI-Powered Student Enrollment ETL Automation

Buildathon / Project Submission Documentation

Organization: Social Eagle AI Academy
Project Type: AI-Powered ETL Automation
Technology: n8n + AI + CSV Processing
Dataset: Student Enrollment Data
Total Records Processed: 50

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1. Executive Summary

This project implements an **AI-powered Student Enrollment ETL Automation pipeline using n8n**.

Social Eagle AI Academy receives student enrollment information periodically through online forms. Such raw data may contain inconsistent information, invalid email addresses, different location formats, and other data-quality issues.

The objective of this project is to automate the complete data-processing workflow using n8n.

The solution:

1. Extracts student enrollment data from a CSV file.
2. Processes every student record individually.
3. Uses AI-assisted processing through the **Phrase AI Response** step.
4. Determines the student's location and email validity.
5. Routes records into appropriate business categories.
6. Generates separate output CSV files.
7. Verifies that every input record has been processed.

The completed workflow successfully processed **all 50 student records** and distributed them into four output categories.

Final Results

Output Category	Records
`chennai_valid_email`	9
`chennai_invalid_email`	7
`other_valid_email`	14
`other_invalid_email`	20
Total	**50**

The reconciliation confirms:

9 + 7 + 14 + 20 = 50

Therefore, all input records were successfully accounted for.

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2. Problem Statement

Student enrollment data collected through Google Forms or similar sources is often not ready for direct business use.

Common problems include:

- Inconsistent student information
- Invalid email addresses
- Different ways of entering locations
- Incorrect or incomplete information
- Manual data-cleaning requirements
- Difficulty separating valid and invalid records
- Repetitive weekly processing

Performing these activities manually can be time-consuming and can introduce human errors.

Business Requirement

Build an automated ETL pipeline using n8n that can:

- Read raw student enrollment data.
- Process student records automatically.
- Apply AI-assisted data validation.
- Identify Chennai and non-Chennai students.
- Identify valid and invalid email records.
- Separate the data into appropriate output files.

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3. Project Objectives

The major objectives of this project are:

3.1 Automated Data Extraction

Read student enrollment records from a CSV input file without manually processing individual records.

3.2 Automated Transformation

Process and prepare each student record for validation and classification.

3.3 AI-Assisted Data Validation

Use the **Phrase AI Response** step to assist in analyzing the student information and determining the required classification.

3.4 Record-Level Processing

Configure the AI processing step with:

Run Once for Each Item

This ensures that each student record is processed individually.

3.5 Automated Classification

Classify every student into one of four categories:

- Chennai + Valid Email
- Chennai + Invalid Email
- Other Location + Valid Email
- Other Location + Invalid Email

3.6 Automated Output Generation

Create separate CSV outputs for each category.

3.7 Data Reconciliation

Verify that the total number of records in all output files equals the number of input records.

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4. High-Level Solution Architecture

The implemented solution follows an ETL architecture:

```
                 RAW STUDENT CSV
                        |
                        v
               +---------------+
               |   n8n Input   |
               | CSV Processing|
               +---------------+
                        |
                        v
               +---------------+
               | Record-Level  |
               |  Processing   |
               +---------------+
                        |
```

```
                 v
        +---------------+
        | Phrase AI     |
        |   Response    |
        +---------------+
                 |
                 v
        +---------------------+
        | Validation /        |
        | Classification      |
        +---------------------+
                 |
       +--------+--------+
       |                 |
    Chennai           Other
       |                 |
  +----+----+       +----+----+
  |         |       |         |
Valid     Invalid  Valid    Invalid
  |         |       |         |
  v         v       v         v
CSV File  CSV File CSV File  CSV File
```

5. ETL Architecture

The project can be divided into three major ETL phases.

5.1 Extract

The raw student enrollment data is provided as a CSV file.

The workflow reads the CSV and loads the student records into n8n.

Input: Raw Student Enrollment CSV

Input Records: 50

5.2 Transform

Each record is processed individually.

The workflow performs AI-assisted analysis using the Phrase AI Response step.

Relevant information such as:

- Student location

- Email information
- Email validity
- Classification information

is processed before routing the record.

The configuration:

Run Once for Each Item

ensures that all 50 records are processed individually.

5.3 Load

After classification, records are routed into four separate output branches.

Each branch generates a corresponding CSV file.

The final output consists of four datasets:

```
chennai_valid_email.csv
chennai_invalid_email.csv
other_valid_email.csv
other_invalid_email.csv
```

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6. Detailed Workflow

Step 1 — Input Student CSV

The workflow starts with the raw student enrollment CSV file.

The CSV contains student enrollment information collected from the source system.

The workflow loads all records into n8n.

Input

50 student records

Step 2 — Process Individual Records

The workflow processes each student record individually.

The Phrase AI Response step is configured as:

Run Once for Each Item

This is important because the AI processing needs to evaluate each student record separately.

Therefore:

```
50 Input Records
       |
       v
50 Individual AI Processing Operations
```

This prevents records from being treated as a single combined dataset.

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7. AI Processing

The project uses an AI-assisted processing step called:

Phrase AI Response

The purpose of this step is to assist with interpreting and classifying the student data.

AI is useful because real-world form data can contain variations that may not always be handled effectively by simple fixed rules.

For example, student location information may have variations in formatting or wording.

The AI processing provides an additional interpretation layer before the records are routed.

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8. Business Classification Logic

After processing, each student record is classified using two major conditions:

Condition 1 — Location

```
Chennai
OR
Other Location
```

Condition 2 — Email Validity

```
Valid Email
OR
Invalid Email
```

Combining these two conditions creates four possible categories.

Classification Rules

```
IF Location = Chennai
AND Email = Valid
        |
        v
chennai_valid_email
```

```
IF Location = Chennai
AND Email = Invalid
        |
        v
chennai_invalid_email
```

```
IF Location != Chennai
AND Email = Valid
        |
        v
other_valid_email
```

```
IF Location != Chennai
AND Email = Invalid
        |
        v
other_invalid_email
```

---

9. Final Output Files

The workflow generates four separate output files.

9.1 Chennai — Valid Email

File:

`chennai_valid_email`

Records: 9

These records represent students whose location is Chennai and whose email information is classified as valid.

9.2 Chennai — Invalid Email

File:

`chennai_invalid_email`

Records: 7

These records represent Chennai students whose email information is classified as invalid.

9.3 Other Location — Valid Email

File:

`other_valid_email`

Records: 14

These records represent students located outside Chennai whose email information is classified as valid.

9.4 Other Location — Invalid Email

File:

`other_invalid_email`

Records: 20

These records represent students located outside Chennai whose email information is classified as invalid.

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10. Final Execution Results

The final workflow produced the following results:

#	Output File / Category	Records	Status
1	`chennai_valid_email`	9	PASS
2	`chennai_invalid_email`	7	PASS
3	`other_valid_email`	14	PASS
4	`other_invalid_email`	20	PASS
	Total	**50**	**PASS**

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11. Data Reconciliation

A critical part of an ETL pipeline is verifying that records are not lost during transformation or routing.

The input contained:

50 records

The four output files contained:

```
Chennai Valid       =  9
Chennai Invalid     =  7
Other Valid         = 14
Other Invalid       = 20
                     ----
Total               = 50
```

Therefore:

Input Records = Output Records

50 = 50

Reconciliation Result

PASS

This confirms that all 50 student records were successfully routed into one of the four output categories.

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12. Testing and Validation

The completed workflow was manually verified after execution.

Test Case 1 — Input Record Count

Expected:

50 records

Actual:

50 records

Result: PASS

Test Case 2 — Individual AI Processing

Configuration:

Run Once for Each Item

Expected:

Each student record processed independently.

Actual:

All 50 records were processed.

Result: PASS

Test Case 3 — Chennai Valid Email

Expected output:

`chennai_valid_email`

Actual:

9 records

Result: PASS

Test Case 4 — Chennai Invalid Email

Expected output:

`chennai_invalid_email`

Actual:

7 records

Result: PASS

Test Case 5 — Other Valid Email

Expected output:

`other_valid_email`

Actual:

14 records

Result: PASS

Test Case 6 — Other Invalid Email

Expected output:

`other_invalid_email`

Actual:

20 records

Result: PASS

Test Case 7 — Final Reconciliation

Expected:

All 50 records accounted for.

Actual:

9 + 7 + 14 + 20 = 50

Result: PASS

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13. Key n8n Concepts Demonstrated

This project demonstrates several important n8n and data-engineering concepts.

13.1 Workflow Automation

The entire data-processing pipeline is implemented as an automated n8n workflow.

13.2 ETL Processing

The project follows:

Extract → Transform → Load

13.3 CSV Processing

The workflow accepts raw CSV data and generates processed CSV outputs.

13.4 Item-by-Item Processing

The workflow demonstrates how n8n can process records individually.

13.5 AI Integration

AI is incorporated into a traditional ETL pipeline for intelligent data interpretation and classification.

13.6 Conditional Branching

The records are routed according to location and email validity.

13.7 Automated File Generation

Each classification branch produces its own output file.

13.8 Data Quality Validation

The workflow separates valid and invalid email records.

13.9 Data Reconciliation

The final output counts are compared against the input count to ensure that no records are lost.

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14. Why AI Was Used

Traditional ETL systems normally depend on predefined rules.

For example:

```
IF city = "Chennai"
THEN Chennai
```

However, real-world data may contain different representations of the same information.

AI can provide an interpretation layer that helps handle less-structured information.

The project therefore combines:

AI + Deterministic ETL Logic

The AI assists with interpretation and classification, while n8n's branching logic ensures predictable final routing.

This combination provides both:

- Intelligent processing
- Controlled business logic

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15. Business Benefits

The solution provides several practical benefits for an organization such as Social Eagle AI Academy.

Reduced Manual Work

Employees do not need to manually inspect every student record.

Improved Data Quality

Invalid email records can be separated from usable records.

Faster Processing

50 records can be processed automatically in a single workflow execution.

Consistent Processing

The same workflow logic is applied to every record.

Easy Segmentation

The output is automatically separated into Chennai/non-Chennai and valid/invalid email groups.

Reusable Automation

The workflow can be reused whenever a new student enrollment file is received.

AI-Powered Data Engineering

The project demonstrates how AI can be integrated into a practical ETL pipeline rather than being used only for chatbot applications.

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16. Scalability

Although the current test dataset contains 50 records, the architecture can be extended for larger datasets.

Possible future scale-up options include:

- Scheduled workflow execution
- Database integration
- Cloud storage
- Batch processing
- Error handling
- Logging
- Notification systems
- Dashboard integration
- Automated weekly processing

The workflow can therefore evolve from a buildathon prototype into a production-oriented data automation solution.

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17. Future Enhancements

The following improvements can be implemented in future versions.

17.1 Duplicate Detection

Detect duplicate student registrations automatically.

17.2 Phone Number Validation

Validate student phone numbers and normalize country codes.

17.3 Additional Email Validation

Implement advanced email validation and domain verification.

17.4 Database Integration

Store cleaned student records in:

- PostgreSQL
- MySQL
- Amazon RDS
- Other enterprise databases

17.5 Cloud Storage

Automatically store processed files in:

- Amazon S3
- Google Cloud Storage
- Azure Blob Storage

17.6 Automated Notifications

Send notifications when processing is completed or when invalid records are detected.

17.7 Scheduling

Configure the workflow to run automatically whenever new weekly enrollment data becomes available.

17.8 Error Handling

Create a dedicated error/rejection branch for records that cannot be processed.

17.9 Monitoring Dashboard

Create a dashboard showing:

- Total records
- Valid records
- Invalid records
- Chennai students
- Other-location students
- Processing failures

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18. Project Outcome

The project successfully demonstrates an end-to-end **AI-powered ETL automation pipeline using n8n**.

The workflow processed:

50 student enrollment records

and successfully classified them into four business categories:

- 9 Chennai + Valid Email
- 7 Chennai + Invalid Email
- 14 Other Location + Valid Email
- 20 Other Location + Invalid Email

All four output files were verified.

The final reconciliation:

9 + 7 + 14 + 20 = 50

confirms that the complete input dataset was processed successfully.

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19. Buildathon Demonstration Flow

During the buildathon presentation, the project can be demonstrated in the following sequence:

Step 1

Show the raw student enrollment CSV.

Explain that the data represents raw enrollment information received from the source system.

Step 2

Open the n8n workflow.

Explain the overall ETL pipeline.

Step 3

Show the individual record processing.

Highlight:

Run Once for Each Item

Explain that every student record is processed independently.

Step 4

Show the Phrase AI Response step.

Explain that AI is used to assist with student data interpretation and classification.

Step 5

Show the branching logic.

Explain:

```
Location
   +
Email Validity
   =
4 Output Categories
```

Step 6

Show the four generated CSV files.

```
chennai_valid_email
chennai_invalid_email
other_valid_email
other_invalid_email
```

Step 7

Show the final record counts.

```
9 + 7 + 14 + 20 = 50
```

Step 8

Conclude with the successful reconciliation.

50 input records → 50 processed records

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20. Key Innovation

The key innovation of this project is the combination of **AI-assisted data processing with workflow-based ETL automation**.

Instead of implementing only traditional rule-based data processing, the solution introduces AI into the transformation stage while retaining deterministic downstream business rules.

The architecture can be summarized as:

Raw Data → AI-Assisted Processing → Business Rules → Automated Outputs

This demonstrates a practical approach to integrating Generative AI into modern data-engineering workflows.

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21. Technology Stack

Technology	Purpose
n8n	Workflow orchestration and automation
AI / Phrase AI Response	AI-assisted data processing
CSV	Input and output data format
ETL Concepts	Data extraction, transformation and loading
Conditional Branching	Business-rule-based routing

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22. Final Project Summary

Project

AI-Powered Student Enrollment ETL Automation

Platform

n8n

Input

Student Enrollment CSV

Input Records

50

AI Processing

Phrase AI Response

Processing Mode

Run Once for Each Item

Output Files

4

Final Distribution

Category	Count
Chennai + Valid Email	9
Chennai + Invalid Email	7
Other + Valid Email	14
Other + Invalid Email	20
Total	**50**

Overall Result

SUCCESSFULLY COMPLETED AND VERIFIED

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23. Conclusion

The **N8N AI-Powered Student Enrollment ETL Automation** project successfully demonstrates how n8n and AI can be combined to automate a real-world data-engineering problem.

The solution takes raw student enrollment data, processes each record individually, applies AI-assisted validation/classification, routes records according to business conditions, and generates clean output datasets.

The successful processing of all **50 records**, with the final distribution of **9, 7, 14, and 20 records**, demonstrates that the workflow is functioning correctly.

The project provides a strong foundation for extending the solution into a production-grade automated student data-processing platform with database integration, cloud storage, scheduling, monitoring, notifications, and advanced data-quality validation.

Final Status: COMPLETED — ALL 50 RECORDS SUCCESSFULLY PROCESSED AND RECONCILED.